

Lecture-20
Mathematics 1 (15B11MA111)

CO [C105.4]

Module: Vector Calculus

Topics to be covered

- **Line Integrals**

- **Examples**

- **Line Integrals Independent of Path**

- **Examples**

- **Assignments**

Reference for the lecture

R.K Jain and S.R.K. Iyenger, “Advanced Engineering Mathematics” fifth edition, Narosa publishing house, 2016.

Vector Integral Calculus

- Vector integral calculus extends the concepts of integral calculus to vector functions.
- It has applications in fluid flow, design of underwater transmission cables, study of satellites.

Line Integrals:

Consider a continuous vector point function $\vec{F}(\vec{r}) = \vec{F}(x, y, z)$ defined at every point of a curve C .

The line integral of $\vec{F}$ along the curve C defined as

$$\int_C \vec{F} \cdot d\vec{r} \quad \text{Or} \quad \int_C \vec{F} \cdot \frac{d\vec{r}}{dt} dt \quad \text{Or} \quad \int_C (F_1 dx + F_2 dy + F_3 dz)$$

The vector $d\vec{r}$ is tangent to the curve, therefore we have

$$d\vec{r} = \hat{t} |d\vec{r}| = \hat{t} ds \quad \int_C \vec{F} \cdot d\vec{r} = \int_C \vec{F} \cdot \hat{t} ds$$

The line integral of the unit tangent vector gives

$$\int_C \hat{t} \cdot d\vec{r} = \int_C \hat{t} \cdot \hat{t} ds = \int_C ds = \text{Length of curve } C.$$

Since there can be infinitely many curves of different lengths joining two points, therefore we conclude:

The line integral depends on the curve.

Applications of Line Integral

If the vector point function represents force acting on body (particle) then the line integral gives the work done in moving the body (particle) along the curve C from point P_1 to point P_2 .

If the vector point function represents the velocity of fluid then the line integral gives **Flow** along the curve.

The line integral in this case is called **Flow integral**.

If curve C is closed (*example a circle or ellipse*) then flow is called **circulation**.

The integral in this case is written as $\oint \vec{F} \cdot d\vec{r}$

If the circulation of a vector point function^c about a closed curve vanishes, then vector is **irrotational**.

Converse is also true i.e.

If vector is irrotational then its circulation about a closed curve vanishes.

The line integral in this case is independent of the path (or curve).

Example: Evaluate the $\int_C \vec{F} \cdot d\vec{r} = \int_C (2ydx - x^2dy)$ where C is the

- (i) line $y = x$ joining two points $(0, 0)$ and $(1,1)$
- (ii) Parabola $y = x^2$ joining two points $(0, 0)$ and $(1,1)$.

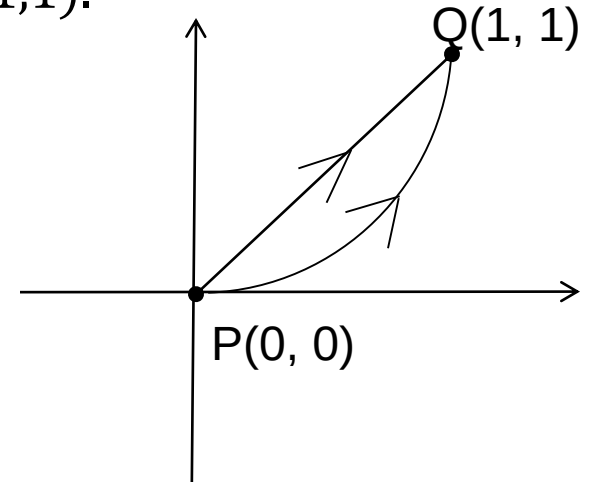
Solution:

$$(i) \int_C \vec{F} \cdot d\vec{r} = \int_C (2ydx - x^2dy)$$

Using parametric form of the equation $y = x = t$ (say)

$$dx = dt, dy = dt$$

$$\begin{aligned} I &= \int_{t=0}^1 (2t dt - t^2 dt) \\ &= \left(t^2 - \frac{t^3}{3} \right) \Big|_0^1 = \frac{2}{3} \end{aligned}$$



$$(ii) \int_C \vec{F} \cdot d\vec{r} = \int_C (2ydx - x^2dy)$$

Using parametric form of the parabola $x = t$ $y = t^2$ and $dx = dt, dy = 2t dt$

$$I = \int_{t=0}^1 (2t^2 dt - 2t^3 dt) = \left(\frac{2t^3}{3} - \frac{2t^4}{4} \right) \Big|_0^1 = \frac{1}{6}$$

Thus in general the line integral depends on the path. But for some special cases it does not depend on the paths.

Example: Calculate $\int_C \vec{F} \cdot d\vec{r}$ where $\vec{F} = xy\hat{i} + x^2y^2\hat{j}$

and C is the quarter circle with center at (0, 0) from (2,0) to (0, 2).

Solution:

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (xy dx + x^2 y^2 dy)$$

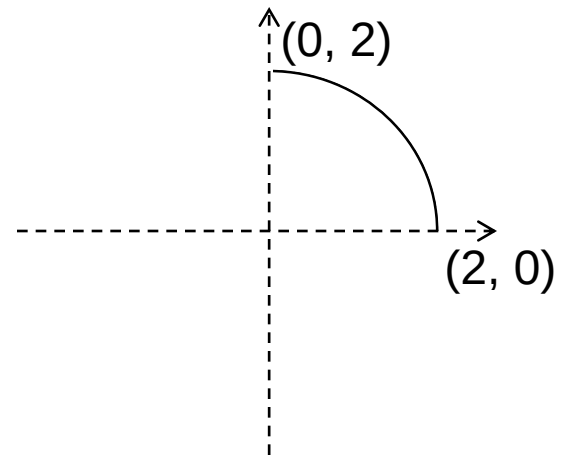
Curve C is part of the circle $x^2 + y^2 = 4$ from (2, 0) to (0,2).

Therefore $y = \sqrt{4 - x^2}$

$$2x dx + 2y dy = 0$$

$$dy = -\frac{x}{y} dx = -\frac{x}{\sqrt{4 - x^2}} dx$$

and x varies from 2 to 0.



$$\begin{aligned}
 \int_C \vec{F} \cdot d\vec{r} &= \int_2^0 \left(x\sqrt{4-x^2}dx + x^2(4-x^2)\left(-\frac{x}{\sqrt{4-x^2}}dx\right) \right) \\
 &= \int_2^0 \left(x\sqrt{4-x^2}dx - x^3\sqrt{(4-x^2)}dx \right) \\
 &= \int_2^0 (1-x^2)x\sqrt{4-x^2}dx
 \end{aligned}$$

$$\text{Let } 4-x^2=t \Rightarrow -2x dx = dt$$

$$\begin{aligned}
 &= \int_0^4 (t-3)\sqrt{t} \frac{dt}{-2} \\
 &= \frac{1}{-2} \left(\frac{t^{5/2}}{5/2} - 3 \frac{t^{3/2}}{3/2} \right) \Big|_0^4 \\
 &= \frac{8}{5}
 \end{aligned}$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_C (xydx + x^2y^2dy)$$

Second way:

Parametric form of quarter circle $x^2 + y^2 = 4$ is $x = 2\cos\theta, y = 2\sin\theta; 0 \leq \theta \leq \frac{\pi}{2}$

$$dx = -2\sin\theta d\theta, \quad dy = 2\cos\theta d\theta;$$

$$\int_C \vec{F} \cdot d\vec{r} = \int_0^{\frac{\pi}{2}} (-8\cos\theta \sin^2\theta + 32\cos^3\theta \sin^2\theta) d\theta$$

Using gamma function

$$= \frac{8}{5}$$

Example: Calculate $\int_C (3y \, dx + 3x \, dy + 2z \, dz)$ where C

is a path from $(0, 0, 0)$ to $(4, 1, 2)$.

Solution: Calculate $\int_{(0,0,0)}^{(4,1,2)} (3y \, dx + 3x \, dy + 2z \, dz)$

$$\int_{(0,0,0)}^{(4,1,2)} (3y \, dx + 3x \, dy + 2z \, dz) = \int_{(0,0,0)}^{(4,1,2)} \vec{F} \cdot d\vec{r}$$

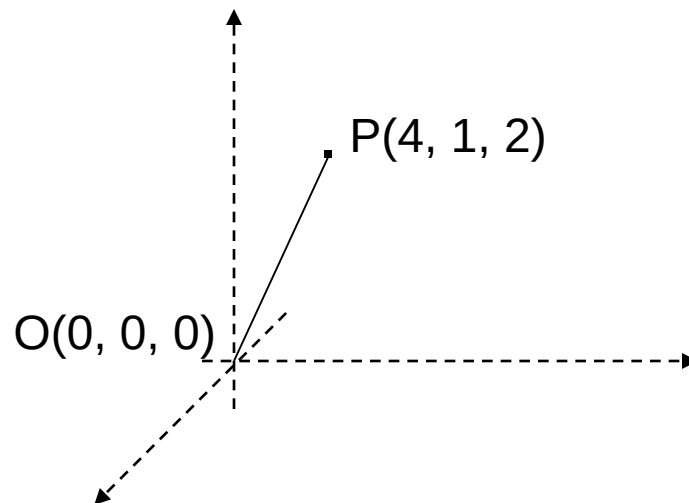
$$\vec{F} = 3y\hat{i} + 3x\hat{j} + 2z\hat{k}$$

We choose the path along the straight line from $(0, 0, 0)$ to $(4, 1, 2)$.

The equation of the line is

$$\frac{x-0}{4-0} = \frac{y-0}{1-0} = \frac{z-0}{2-0}$$

$$\frac{x}{4} = \frac{y}{1} = \frac{z}{2}$$



Using these equations, we express the line integral in terms of x .

We have $x = 4y$, $dx = 4dy$, $z = \frac{1}{2}x$, $dz = \frac{1}{2}dx$

and x varies from 0 to 4.

Therefore

$$\begin{aligned}
& \int_{(0,0,0)}^{(4,1,2)} (3y \, dx + 3x \, dy + 2z \, dz) \\
&= \int_0^4 \left(3\left[\frac{x}{4}\right] dx + 3x\left[\frac{1}{4} dx\right] + 2\left[\frac{x}{2}\right] \frac{1}{2} dx \right) \\
&= \int_0^4 \left(\frac{3}{4}x + \frac{3}{4}x + \frac{x}{2} \right) dx \\
&= \int_0^4 2x \, dx \\
&= \left| x^2 \right|_0^4 = 16
\end{aligned}$$

Example: A fluid velocity field is $\vec{F} = x\hat{i} + z\hat{j} + y\hat{k}$. Find the flow along the helix

$$\vec{r}(t) = (\cos t)\hat{i} + (\sin t)\hat{j} + t\hat{k}, \quad 0 \leq t \leq \pi.$$

Solution: On the curve

$$\vec{F} = x\hat{i} + z\hat{j} + y\hat{k} = (\cos t)\hat{i} + t\hat{j} + (\sin t)\hat{k}$$

$$\frac{d\vec{r}}{dt} = (-\sin t)\hat{i} + (\cos t)\hat{j} + \hat{k}$$

$$\vec{F} \cdot \frac{d\vec{r}}{dt} = -\cos t \sin t + t \cos t + \sin t$$

$$\begin{aligned} \text{Flow} &= \int_C \vec{F} \cdot \frac{d\vec{r}}{dt} dt = \int_0^\pi [-\cos t \sin t + t \cos t + \sin t] dt \\ &= \left| \frac{\cos^2 t}{2} + t \sin t \right|_0^\pi = \left(\frac{1}{2} + 0 \right) - \left(\frac{1}{2} - 0 \right) = 0 \end{aligned}$$

Example: Find the circulation of the field $\vec{F} = (x - y)\hat{i} + x\hat{j}$ around the circle

$$\vec{r}(t) = (\cos t)\hat{i} + (\sin t)\hat{j}, \quad 0 \leq t \leq 2\pi.$$

Solution: On the circle

$$\vec{F} = (x - y)\hat{i} + x\hat{j} = (\cos t - \sin t)\hat{i} + (\cos t)\hat{j}$$

$$\frac{d\vec{r}}{dt} = (-\sin t)\hat{i} + (\cos t)\hat{j}$$

$$\vec{F} \cdot \frac{d\vec{r}}{dt} = -(\cos t - \sin t)\sin t + \cos^2 t = -\cos t \sin t + 1$$

$$\text{Circulation} = \int_C \vec{F} \cdot \frac{d\vec{r}}{dt} dt = \int_0^{2\pi} [1 - \cos t \sin t] dt$$

$$= \left| t + \frac{\cos^2 t}{2} \right|_0^{2\pi} = \left(2\pi + \frac{1}{2} \right) - \left(0 + \frac{1}{2} \right) = 2\pi$$

Example: Find the work done by the gradient of $f(x, y) = (x + y)^2$ counterclockwise around the circle $x^2 + y^2 = 4$ from $(2, 0)$ to itself.

Solution:

$$\begin{aligned} \text{grad } f &= \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} && \text{[2-D case]} \\ &= \hat{i} 2(x + y) + \hat{j} 2(x + y) = 2(x + y)(\hat{i} + \hat{j}) \end{aligned}$$

On the circle $x = 2 \cos \theta$, $y = 2 \sin \theta$, $0 \leq \theta \leq 2\pi$.

$$\vec{r} = (2 \cos \theta) \hat{i} + (2 \sin \theta) \hat{j} \quad d\vec{r} = (-2 \sin \theta d\theta) \hat{i} + (2 \cos \theta d\theta) \hat{j}$$

$$\text{grad } f = 2(2 \cos \theta + 2 \sin \theta)(\hat{i} + \hat{j}) = 4(\cos \theta + \sin \theta)(\hat{i} + \hat{j})$$

$$\text{grad } f \cdot d\vec{r} = 4(\cos \theta + \sin \theta)(-2 \sin \theta + 2 \cos \theta) d\theta$$

$$\text{grad } f \cdot d\vec{r} = 8(\cos^2 \theta - \sin^2 \theta) d\theta = 8 \cos 2\theta d\theta$$

$$\text{Work} = \int_0^{2\pi} 8 \cos 2\theta d\theta = \left| 4 \sin 2\theta \right|_0^{2\pi} = 0$$

Independence of path; conservative field and scalar potential

If the force is conservative then $\nabla \times \vec{F} = 0 \Rightarrow \vec{F} = \nabla f$ where f is the scalar potential and the line integral is independent of path from P_1 to P_2 .

$$\begin{aligned} \int_{P_1}^{P_2} \vec{F} \cdot d\vec{r} &= \int_{P_1}^{P_2} \nabla f \cdot d\vec{r} = \int_{P_1}^{P_2} \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy + \frac{\partial f}{\partial z} dz = \int_{P_1}^{P_2} df \\ &= f(P_2) - f(P_1) \end{aligned}$$

Thus when force is conservative then the line integral is independent of paths i.e. depends only on the end points P_1 and P_2 and not on the path joining them.

- **Result 1:** The work done in a conservative force field in moving a particle from P_1 to P_2 is independent of path joining P_1 and P_2 . In such cases a scalar potential f exists such that force field $= \nabla f$ and work done $= f(P_2) - f(P_1)$.
- **Result 2:** In a conservative field $\vec{F}$, work done along any closed curve C is always zero

$$\oint_C \vec{F} \cdot d\vec{r} = 0$$

while in case of irrotational field it is called circulation which will be zero.

Example: Show that $\int_C (xy^2z^2dx + x^2yz^2dy + (x^2y^2z + 2z)dz)$ is independent of path of integration from P(1,0,1) to Q(0,2,2). Evaluate the integral also.

Solution:

The line integral will be independent of path when $\nabla \times \vec{F} = 0$.

So first find $\text{curl} \vec{F}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2z^2 & x^2yz^2 & (x^2y^2z + 2z) \end{vmatrix}$$

$$\nabla \times \vec{F} = \dots = 0$$

Hence the given vector field is conservative and can be expressed as the gradient of a scalar function $f(x, y, z)$. i.e. $\vec{F} = \text{grad}(f)$.

Hence

$$\nabla f = \hat{i} \frac{\partial f}{\partial x} + \hat{j} \frac{\partial f}{\partial y} + \hat{k} \frac{\partial f}{\partial z} = \vec{F} = xyz(yz\hat{i} + xz\hat{j}) + (x^2y^2z + 2z)\hat{k}$$

Comparing, we get

$$\frac{\partial f}{\partial x} = xy^2z^2, \frac{\partial f}{\partial y} = x^2yz^2, \frac{\partial f}{\partial z} = x^2y^2z + 2z.$$

Integrating the first equation, we get $f(x, y, z) = \frac{1}{2}x^2y^2z^2 + g(y, z)$.

Substituting in the second equation, we get

$$\frac{\partial f}{\partial y} = x^2yz^2 = x^2yz^2 + \frac{\partial g}{\partial y} \text{ or } \frac{\partial g}{\partial y} = 0 \text{ or } g = g(z).$$

Substituting in the third equation, we get

$$\frac{\partial f}{\partial z} = x^2y^2z + 2z = x^2y^2z + \frac{dg}{dz} \quad \text{or} \quad \frac{dg}{dz} = 2z \quad \text{or} \quad g = z^2 + c,$$

where c is a constant.

Hence, $f(x, y, z) = \frac{1}{2}x^2y^2z^2 + z^2 + c$. Therefore, there exist a scalar function $f(x, y, z)$ such that $\nabla f = \bar{F}$ and the vector field is conservative and line integral is independent of path.

Hence

$$\begin{aligned}\int_C (xy^2z^2dx + x^2yz^2dy + (x^2y^2z + 2z)dz) &= \int_P^Q df \\ &= [f(x, y, z)]_P^Q \\ &= f(0, 2, 2) - f(1, 0, 1) \\ &= (4 + c) - (1 + c) \\ &= 3\end{aligned}$$

Assignment

1. Show that the vector field $\vec{F} = 2x(y^2 + z^3)\hat{i} + (2x^2y + 1)\hat{j} + 3x^2z^2\hat{k}$ is conservative. Find its scalar potential and work done in moving a particle from $(-1, 2, 1)$ to $(2, 3, 4)$. Ans. 288

Thank you